

SALALAH, Oman

WMO: 413160

Lat: 17.039N

Lon: 54.091E

Elev: 22

StdP: 101.06

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 17.9	(c) 18.9	(d) -7.1	(e) 2.1	(f) 23.8	(g) -4.0	(h) 2.7	(i) 25.2	(j) 11.7	(k) 22.3	(l) 10.4	(m) 23.3	(n) 3.1	(o) 350	(p) 0.353

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 5	(b) 5.0	(c) 33.8	(d) 21.6	(e) 32.9	(f) 23.8	(g) 32.1	(h) 24.5	(i) 28.1	(j) 30.6	(k) 27.6	(l) 30.3	(m) 27.2	(n) 30.2	(o) 5.6	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	23.0	30.0	26.9	22.7	29.8	26.2	21.7	29.4	90.5	31.3	88.6	30.4	86.8	30.4	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.2	8.0	7.1	DB	15.7	39.1	1.1	3.0	15.0	41.2	14.3	43.0	13.7	44.7	12.9	46.9	
			WB	6.8	28.5	1.3	0.9	5.9	29.1	5.1	29.6	4.4	30.1	3.5	30.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	23.3	24.0	26.0	28.2	29.8	29.6	26.5	25.5	27.0	27.1	26.6	25.0
	DBStd	2.33	1.35	1.65	1.60	1.29	1.05	1.21	1.38	1.06	1.01	1.29	1.36	1.51
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6047	413	392	497	546	615	588	513	481	508	531	499	464
	CDD18.3	3006	154	158	239	296	357	338	254	223	258	273	249	206
	CDH23.3	30092	863	1083	2274	3580	4772	4334	2221	1464	2433	2937	2535	1596
Wind	WSAvg	3.2	3.4	3.1	2.9	3.0	3.5	3.9	3.2	2.8	3.1	2.5	2.8	3.8
	PrecAvg	73	1	1	6	11	2	3	23	22	3	5	2	0
	PrecMax	172	10	11	91	151	42	15	61	63	15	69	44	12
	PrecMin	21	0	0	0	0	0	0	0	3	0	0	0	0
	PrecStd	34	2	2	16	29	7	3	12	12	4	12	7	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.7	31.9	34.4	36.1	34.6	33.8	31.1	29.2	30.9	34.9	34.2	32.0
		MCWB	15.8	16.4	17.1	18.2	23.1	27.5	26.7	25.3	25.1	19.5	18.2	17.1
	2%	DB	28.2	29.9	32.1	32.8	33.1	32.9	30.1	28.2	30.1	32.4	32.2	30.2
		MCWB	16.5	16.3	17.5	21.7	26.1	27.2	26.3	24.9	24.6	20.8	18.3	17.3
	5%	DB	27.2	28.2	30.7	31.8	32.5	32.1	29.1	27.8	29.6	31.2	31.0	29.2
		MCWB	18.1	18.4	19.3	23.7	26.2	26.9	25.9	24.8	24.4	22.4	19.8	18.1
	10%	DB	26.8	27.2	29.2	31.0	32.0	31.8	28.4	27.0	29.0	30.2	30.1	28.2
		MCWB	18.5	19.4	21.2	24.2	26.1	26.8	25.6	24.6	24.2	23.1	21.1	19.1
	0.4%	WB	22.5	23.7	25.2	27.2	28.4	29.0	27.5	26.2	26.3	26.1	25.5	23.9
		MCDB	26.1	26.9	28.3	30.3	31.8	31.6	29.3	27.7	28.4	29.1	28.9	27.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.8	22.7	24.5	26.5	27.7	28.4	26.8	25.8	25.9	25.5	24.7	22.7
		MCDB	25.7	26.5	28.0	30.0	30.9	30.9	28.6	27.2	28.0	28.9	28.6	26.8
	5%	WB	21.1	22.2	24.1	26.0	27.2	27.9	26.4	25.5	25.6	25.1	24.0	22.1
		MCDB	25.4	26.2	27.8	29.7	30.7	30.4	28.2	26.9	27.8	28.7	28.3	26.7
	10%	WB	20.5	21.5	23.6	25.5	26.9	27.6	26.0	25.1	25.2	24.6	23.4	21.5
		MCDB	25.1	25.8	27.5	29.5	30.5	30.2	27.8	26.4	27.5	28.5	28.0	26.5
Mean Daily Temperature Range	5% DB	MDBR	7.5	7.5	7.3	6.2	5.0	3.7	2.8	2.7	4.1	7.2	7.9	7.6
		MCDBR	8.0	8.6	10.0	7.6	5.3	4.3	3.5	3.6	4.7	8.8	9.2	8.1
		MCWBR	5.3	6.2	7.6	5.5	2.7	2.0	1.5	1.4	2.2	6.1	6.3	5.2
	5% WB	MCDBR	6.7	6.6	5.9	5.6	4.9	3.9	3.2	3.0	3.9	6.2	7.2	6.9
		MCWBR	4.6	4.7	4.1	3.5	2.6	2.0	1.5	1.5	2.0	4.5	5.3	4.8
Clear-Sky Solar Irradiance	taub		0.420	0.443	0.495	0.527	0.618	0.791	0.834	0.769	0.620	0.459	0.438	0.421
	taud		2.180	2.101	1.962	1.921	1.738	1.461	1.428	1.528	1.756	2.117	2.161	2.176
	Ebn at Noon		861	859	825	798	719	598	574	618	715	833	835	846
	Edh at Noon		141	160	189	198	234	306	317	289	229	155	142	138
All-Sky Solar Radiation	RadAvg		5.45	6.25	6.95	7.36	7.37	6.61	5.67	5.75	6.29	6.41	5.60	5.22
	RadStd		0.14	0.16	0.24	0.20	0.25	0.32	0.30	0.19	0.25	0.16	0.20	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-2.05	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.15	N/A	-1.52	N/A	N/A	-0.30	N/A	N/A	N/A	N/A

Nomenclature: See separate page